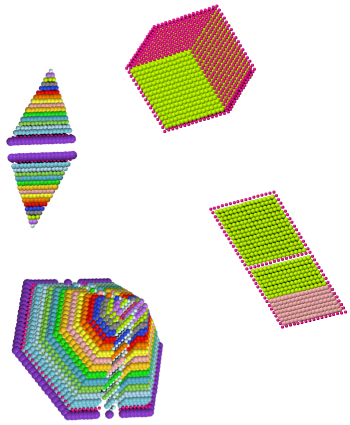
Representation and Construction of Perfects

An Illustrative Exposition of Fermat's Last Theorem and Beal's Conjecture



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Abstract

Author's Note

Keeping things as simple as possible, elliptic curves and cyclotomatic numbers are avoided. This article does not require advanced mathematics to understand. This article does not make reference to any other work. Representation of perfects and the construction presented show solution. Both conjectures are solved by the same approach. One explanation. The conjectures are put into a form which show that the operation (remove or blend) on two perfects defy exact reconstruction of a perfect. The *proof* is simply from the definition of perfects. The mechanism is derived from a *faithful representation* describing all cases of the constructive model. The mechanism shown demonstrates the impossibility of construction within the constraints. The constraint of having symmetric identical sources. During construction balance must be maintained. When the perfect is constructed one layer at a time, from one source, placing the apex is unconstrained. The mechanism's constraint is a property of bilateral symmetry (bls). For a formation to have bls, the formation is required to have an apex and left/right/mirror symmetric sections flanking center line. The center is a linear arrangement of unit points. The center provides the notion of dimension. The fulcrum is the cornerstone of the constructible object. The apex is the last point placed.

This is the only fact needed for *theorem.* Having two or more identical symmetric sources with which to form the perfect constrains construction to two or more layers at a time which results in the exclusion of the apex. To place apex would require having an imbalance of points (ie $weight \bmod sources \neq 0$) with $\mathrm{L} \approx \mathrm{R}$.

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1 Nomenclature

Perfect: Δ, an object with whole number of components and a whole dimensional value which is the nth-order root of the objects measure.
Dimension: The notion of size.
Order: The degree of the dimensional value.
Measure: v. To count the components. n. The count of components.
Point: The smallest unit comprising the components.

Perfects are not defined by length. The notion of distance is useless. Perfects do not have length, area or volume. Perfects are a countable arrangements of components. The components are assigned a *weight*. The value of the measure of the perfect is the total *weight*.

2 Problems

$$\iota \begin{cases} A^2 + B^2 = C^2 & \text{Pythagorean Theorem} \\ A^N + B^N = C^N & \text{Fermat's Last Theorem} \\ A^X + B^Y = C^Z & \text{Beal's Conjecture} \end{cases}$$

3 Computational Algorithm

$$H_\Delta = \begin{cases} 0 & \theta \text{ zero,} \\ \Delta(\theta, \beta, \delta) & \theta,\ \beta \text{ and } \delta \text{ whole.} \end{cases}$$

$$\Delta(\theta, \beta, \delta) = (\beta + \delta)\Delta(\theta - 1, \beta, \delta) + \delta \sum_{r=1}^{\beta-1} \Delta(\theta - 1, r, \delta)$$

$$P^\theta - P^{(\theta-2)} = P^{(\theta-2)}(P^2 - 1)$$

$$C_\theta = P^\theta - P^{(\theta-2)}$$

$$C_\theta \quad \text{mod } 6 \equiv 0$$

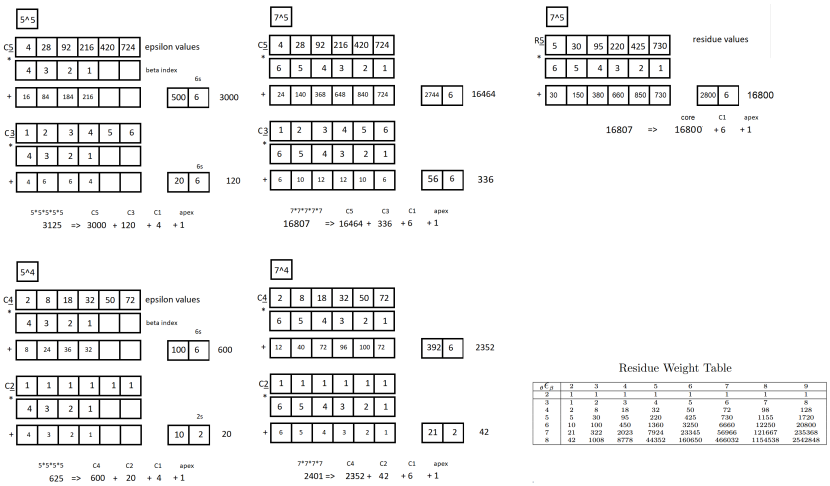
$$C_\theta \quad has \quad six \quad identical \quad parts \quad when \quad \theta > 2$$

$$S_\theta = \Delta(\theta, \beta, \delta) - \Delta((\theta - 2), \beta, \delta)$$

$$\epsilon_\psi = ((\Delta(\theta, \beta + 1, \delta) - \Delta(\theta - 2, \beta + 1, \delta)) - (\Delta(\theta, \beta, \delta) - \Delta(\theta - 2, \beta, \delta))$$

$$\epsilon_\theta = \begin{cases} 1 & \theta = 2 \text{ (flat)}, \\ \epsilon_\psi / 6 & \theta > 2 \ . \end{cases}$$

Equations shown here define the $H_p^y e$ mechanism. See Tables 1-3. The mechanism generates the complete set of differences for perfects of all orders. The difference of two perfects is an additive method. The recursive function uses only addition. Perfects are formed from the sub-polynomial clusters (C_θ). The cluster is shown to have a 6 symmetry(6s). The epsilon weights ϵ_ψ are 6s characteristics of a layer which define the weight of the rings in the cluster.



$n \backslash \epsilon_\theta$	2	3	4	5	6	7	8	9
2	1	1	1	1	1	1	1	1
3	1	2	3	4	5	6	7	8
4	2	8	18	32	50	72	98	128
5	5	30	95	220	425	730	1155	1720
6	10	100	450	1360	3250	6660	12250	20800
7	21	322	2023	7924	23345	56966	121667	235368
8	42	1008	8778	44352	160650	466032	1154538	2542848

Figure 1: sub-polynomial clusters C_θ all cluster have same form. all solid 3d hexagonal pyramid with six wedges which comprise the ΣC_n residue

4 Representation and Construction

The perfect can be represented as a triangular arrangement of points. See Fig 2. Each point has a unit weight for a total *weight* of $N^2 = 1 + 3 + 5 + ... + (2N - 1)$. The dimension or *height* of the perfect is the count of points in the center ie the ϵ_0 column. An even perfect has a plane of symmetry about the C_2 center. An odd perfect has axial symmetry around the C_1 center. See Fig 3 and 4. The C_2 cluster is 2s around the C_1

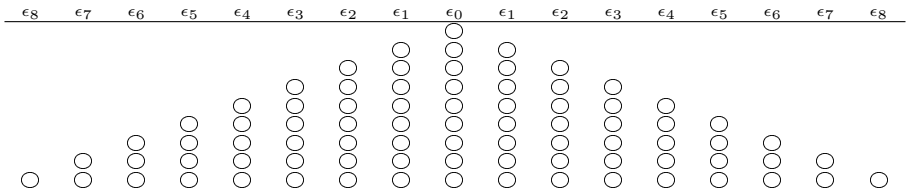


Figure 2: even center C_2

ϵ_0	ϵ_1	ϵ_2	ϵ_3	ϵ_4	ϵ_5	ϵ_6	ϵ_7	ϵ_8
apex								
α_0	ω_1							
α_0	ω_1	ω_2						
α_0	ω_1	ω_2	ω_3					
α_0	ω_1	ω_2	ω_3	ω_4				
α_0	ω_1	ω_2	ω_3	ω_4	ω_5			
α_0	ω_1	ω_2	ω_3	ω_4	ω_5	ω_6		
α_0	ω_1	ω_2	ω_3	ω_4	ω_5	ω_6	ω_7	
α_0	ω_1	ω_2	ω_3	ω_4	ω_5	ω_6	ω_7	ω_8

Figure 3: Odd order wedge $\alpha_0 center$

center for bls. The odd orders are transformed into the same bilateral representation. Each 6s 3d wedge is represented as two 3p ϵ units around the C_1 center.

From the diagrams it can be seen that the difference operation is two sided and produces two symmetric parts which are avialable for rearrangement into the perfect $\Delta_A = \Delta_C - \Delta_B$. See Fig 5. There are three obstacles one must hurdle in order to rearrange points after the difference operation. First the 6s core pieces are formed from 3p components and then the center must be formed from 1p units and finally an apex is placed. One stipulates to the first two of these conditions. Constructing the center requires that ϵ units are decomposed into 1p units. The constraint of left and right sources force the center to be even. The newly forming object must be built two layers at a time. The apex can not be placed atop the ϵ^0 column. See Fig 6. A core can be formed for some values of the triple A, B and C. An even center may be placed, however infrequent. But finally to place the apex is a challenge. 3p6s Core + Even center + Apex would be an odd number of components. QED.

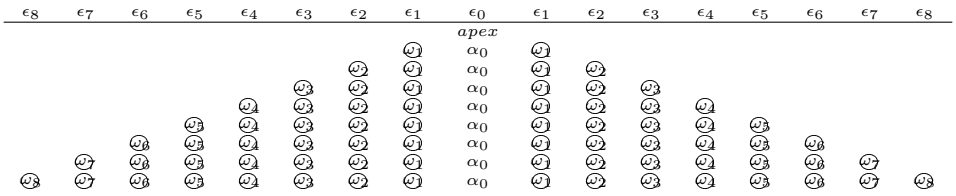


Figure 4: flat 2d bilateral

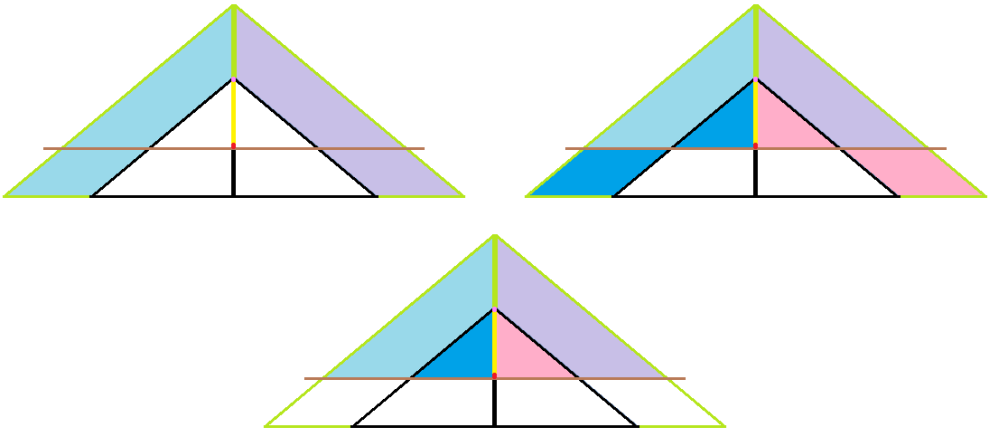


Figure 5: Difference of two perfects. In the diagram on the left, the large (green) triangle is C^2 and the small (black) triangle is B^2. The diagram on the right shows the *points* which are reformed into another perfect in the vacancy left by the difference operation. A^2 is the the colored portion of the center diagram.

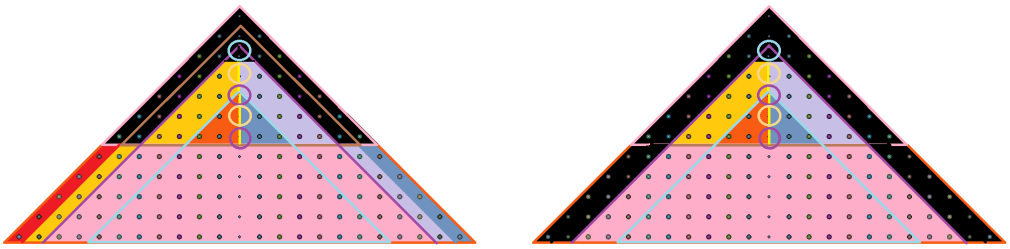


Figure 6: Another equivalent difference of two perfects. Smaller triangle B^N is shaved off the top of C^N with the lower outside portions being used to form the perfect A^N. There are four perfects involved. A^N, B^N and C^N along with the perfect $A^N - (C^N - B^N)$ being formed above the difference. It is readily seen that the center of the perfect being formed on the top of the difference is always even. Placing the apex is not possible without creating an imbalance in the formation. The methods can be used for all orders. It shows for orders above two that it is impossible to rearrange the components into a perfect.

δ 1	0	1	2	3	4	5	6	7	8
1	1	1	1	1	1	1	1	1	1
2	1	3	5	7	9	11	13	15	17
3	1	7	19	37	61	91	127	169	217
4	1	15	65	175	369	671	1105	1695	2465
5	1	31	211	781	2101	4651	9031	15961	26281
6	1	63	665	3367	11529	31031	70993	144495	269297
7	1	127	2059	14197	61741	201811	543607	1273609	2685817
8	1	255	6305	58975	325089	1288991	4085185	11012415	26269505

Table 1: Layer Table
the difference of perfects

${}_\theta\epsilon_1$	2	3	4	5	6	7	8	9
2	1	1	1	1	1	1	1	1
3	1	2	3	4	5	6	7	8
4	2	8	18	32	50	72	98	128
5	4	28	92	216	420	724	1148	1712
6	8	92	432	1328	3200	6588	12152	20672
7	16	292	1928	7704	22920	56236	120512	233648
8	32	908	8328	42992	157400	459372	1142288	2522048

Table 2: Epsilon Characteristic Weight Table
single cluster weight units for rings

${}_{\theta}\epsilon_{\beta}$	2	3	4	5	6	7	8	9
2	1	1	1	1	1	1	1	1
3	1	2	3	4	5	6	7	8
4	2	8	18	32	50	72	98	128
5	5	30	95	220	425	730	1155	1720
6	10	100	450	1360	3250	6660	12250	20800
7	21	322	2023	7924	23345	56966	121667	235368
8	42	1008	8778	44352	160650	466032	1154538	2542848

Table 3: Residue Weight Table
all clusters : sum of epsilon weights that form core